

Assignment 3

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1 Introduction

The objective of this assignment is to simulate the diffusion of information through a social network where different types of actors coexist. The network represents interactions between anime entries extracted from the MyAnimeList dataset. Each node represents an anime, while edges represent similarity relationships based on shared genres.

Three different kinds of information are considered during the simulation:

- a reliable message propagated by Verifiers;
- a mutated message generated by Unintentional Spreaders;
- fake information introduced by Social Bots.

The objective is to analyze how different node behaviors influence the diffusion process and how misinformation propagates through the network.

2 Network Construction

The network was generated from the filtered anime dataset used in the previous assignments.

Each anime is represented by its identifier (`mal_id`) and title. An undirected edge is created between two anime when they share at least two genres.

The graph construction rule is:

- Nodes: anime entries.
- Edges: two anime share at least two genres.
- Edge type: undirected.

The final network contains:

Metric	Value
Nodes	3579
Edges	780393
Average degree	436.10
Density	0.1219

Table 1: Main characteristics of the anime similarity network.

A fixed random seed (42) was used to guarantee reproducibility.

3 Actor Roles

The simulation considers four different types of nodes.

3.1 Verifiers

Verifiers represent reliable information sources. They initially spread the correct message

$$M_{true} = Low$$

and never change their state during the simulation.

3.2 Unintentional Spreaders

These nodes accidentally modify the information they receive. Whenever they adopt the correct message, they transform it into

$$M_{mutated} = Moderate$$

before forwarding it.

3.3 Normal Nodes

Normal nodes follow a threshold model.

A node adopts the message supported by the majority of its neighbors if

$$\frac{\text{neighborssupportingthemessage}}{\text{totalneighbors}} \geq \text{threshold}.$$

Otherwise it remains inactive.

3.4 Social Bots

The second part of the assignment introduces malicious Social Bots.

Bots are selected according to betweenness centrality because these nodes are expected to control important communication paths inside the network.

Whenever activated, they replace any received message with

$$M_{fake} = Fake$$

and propagate it to their neighbors.

4 Diffusion Model

The spreading process evolves in discrete iterations.

At every iteration:

1. Verifiers keep broadcasting the true message.
2. Unintentional nodes may transform the true message into the mutated one.
3. Normal nodes adopt the majority message if the threshold condition is satisfied.
4. Bots replace every received message with fake information.

The simulation stops when no node changes state or after twenty iterations.

5 Experimental Setup

The first experiment was performed using:

- 200 Verifiers;
- 10 Unintentional Spreaders;
- threshold = 0.01.

The second experiment introduced:

- 3 Social Bots;
- bot placement based on betweenness centrality.

To study the phase transition, the number of Unintentional nodes was progressively increased from 0 to 40. For every configuration, 10 independent simulations were executed and the average percentage of mutated messages was computed.

Two placement strategies were compared:

- random selection;
- strategic selection based on betweenness centrality.

6 Results

6.1 Diffusion Without Bots

The final message distribution is reported in Table 2.

Message	Percentage
True	80.53%
Mutated	0.22%
None	19.25%

Table 2: Final message distribution without Social Bots.

The correct information reaches more than 80% of the network. Only eight nodes adopt the mutated message, showing that randomly placed Unintentional Spreaders have a very limited impact on the diffusion process. Approximately 19% of the nodes remain inactive because the threshold condition is not satisfied.

6.2 Diffusion With Bots

After introducing three strategically positioned Social Bots, the final distribution becomes:

Message	Percentage
True	5.62%
Mutated	0.00%
Fake	74.74%
None	19.64%

Table 3: Final message distribution with Social Bots.

The introduction of only three bots dramatically changes the spreading process. Almost three quarters of the network adopt the fake message, while the proportion of nodes receiving the correct information decreases to about 6%. The percentage of inactive nodes remains almost unchanged, indicating that bots mainly replace the propagated content rather than increasing the overall reach of the diffusion.

7 Phase Transition Analysis

The influence of Unintentional Spreaders was analyzed by progressively increasing their number from 0 to 40.

Two different placement strategies were compared:

- random placement;
- strategic placement based on betweenness centrality.

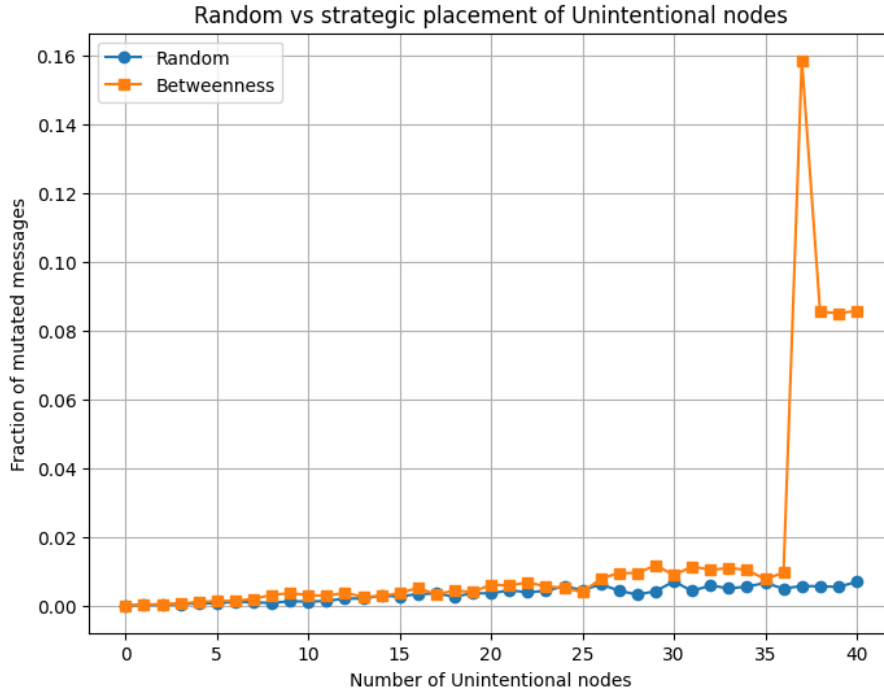


Figure 1: Comparison between random and strategic placement of Unintentional Spreaders.

Random placement produces only a gradual increase in the percentage of mutated messages. In contrast, the strategic placement based on betweenness centrality exhibits a sharp increase around 37 Unintentional nodes. This behaviour suggests the presence of

a phase transition, where a relatively small increase in strategically positioned unreliable actors produces a much larger diffusion of corrupted information.

Discussion

The comparison between random and strategic placement shows that the random strategy produces only a gradual increase in the percentage of mutated messages. In contrast, the betweenness-based strategy exhibits a sudden increase around 37 Unintentional nodes. This behaviour suggests the presence of a phase transition. Nodes with high betweenness centrality act as bridges between different regions of the network. As more of these bridge nodes become Unintentional Spreaders, the mutated message can suddenly propagate from one densely connected region to another, producing a rapid increase in misinformation. Once a sufficient number of strategic bridge nodes is reached, even a small increase in their number can trigger a much larger cascade of corrupted information.

8 Network Visualization

The final network states were visualized using a spring layout.

Node colors represent:

- Green: Verifiers.
- Yellow: Unintentional Spreaders.
- Red: Social Bots.
- Blue: True message.
- Orange: Mutated message.
- Purple: Fake message.
- Gray: Inactive nodes.

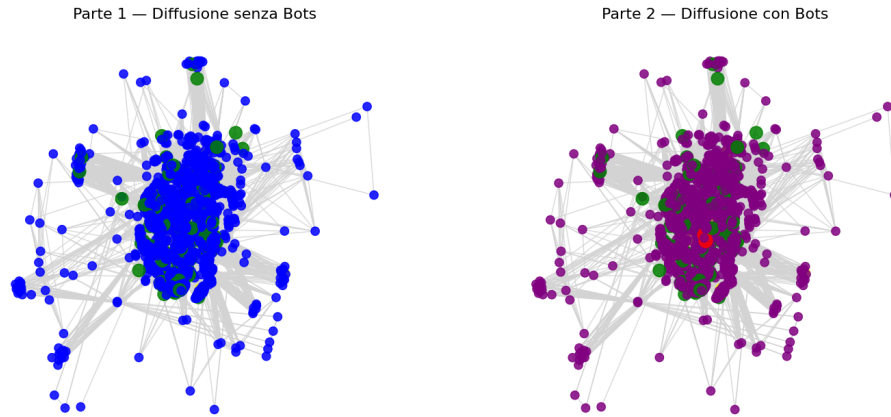


Figure 2: Comparison between the diffusion process without and with Social Bots.

Without bots, most active nodes propagate the true information. After introducing three strategically positioned bots, the same network structure is preserved, but most active nodes switch to the fake message, highlighting the disruptive influence of highly central malicious actors.

9 Technical Decisions

Several implementation choices were made to keep the simulations both realistic and computationally feasible.

- A fixed random seed (42) was used to ensure reproducibility.
- The diffusion process follows a threshold model, where nodes adopt the majority message only if the threshold condition is satisfied.
- Bots and strategically placed Unintentional Spreaders were selected using betweenness centrality because bridge nodes can influence information flow across different parts of the network.

- For computational efficiency, an approximate betweenness centrality ($k = 200$) was used instead of the exact computation, significantly reducing the execution time while preserving the ranking of the most central nodes.
- Neighbor lists were precomputed before the simulations to avoid repeatedly scanning the graph during every iteration, reducing the computational cost of the spreading process.
- The phase transition experiment was repeated ten times for each configuration and the average result was reported in order to reduce the effect of randomness.

10 Conclusion

The experiments show that both network topology and node behaviour strongly influence information diffusion.

Randomly distributed Unintentional Spreaders produce only limited misinformation, whereas strategically positioned nodes have a much larger impact on the spreading process.

The introduction of only three Social Bots is sufficient to replace the true message with fake information over most of the network, demonstrating the importance of central nodes in controlling information propagation.

Overall, the simulations highlight how heterogeneous actors and network structure jointly determine the robustness of information diffusion in complex networks.